

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	XXXXXXX
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the FinRelief AI codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation throughout the project	Yes	Proper formatting followed
2	Proper File Structure	Frontend and backend organized into separate folders	Yes	React and FastAPI folders maintained
3	Meaningful Variable Names	Variables clearly represent their purpose	Yes	Easy to understand and maintain
4	Function / Method Names	Functions use descriptive names	Yes	Improves readability
5	Code Comments	Important functions are documented with comments	Partial	Core modules documented
6	Modular Design	Code divided into reusable modules and components	Yes	Separate authentication, loans, AI, and dashboard modules
7	No Redundant Code	Duplicate code minimized through reusable functions	Yes	Common utilities reused
8	Error Handling	Proper exception handling and input validation	Yes	Backend validates requests and returns meaningful errors

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Authentication Module	FastAPI, JWT	Login, Registration, Profile	High
2	Loan Management Module	Python, FastAPI	Add, Edit, Delete, View Loans	High
3	AI Recommendation Engine	Python, Gemini API	Financial Analysis & Debt Settlement	High
4	Dashboard Components	React.js	Dashboard, Reports, Analytics	High
5	Database Module	SQLite, SQLAlchemy	All backend operations	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
6	API Service	Axios	Frontend API communication	High
7	Report Generator	Python	Financial Summary & Reports	Medium
8	Profile Management	React + FastAPI	User Profile and Settings	Medium